UNIVERSITY OF TORONTO

STA2112 — Supplementary Derivations

Fall 2026

Optional practice accompanying Homework 1; no submission required

Last name: _____ First name: _____

Student number: _____

Instructions

Show your reasoning and intermediate steps, and label all parts clearly. The STA2112 syllabus policies apply. These exercises provide optional practice behind the formulas in lectures. They carry no Homework 1 marks or bonus points and are not prerequisites for the main assignment. Use ordinary differentiation, algebra, and the facts explicitly given below; no additional optimization theory is assumed. Questions 2, 3, and 5 are the optional derivations referenced in Lecture 3.

1. **Ridge and lasso: penalties versus constraints.**

   Assume the predictors and response have been centered, so no intercept is needed. Let $X \in \mathbb{R}^{n\times p}$, $y \in \mathbb{R}^n$, and $L(\beta) = \frac{1}{2}\|y - X\beta\|_2^2$. For ridge use $P(\beta) = \|\beta\|_2^2$; for lasso use $P(\beta) = \|\beta\|_1 = \sum_{j=1}^p |\beta_j|$. Compare the penalized and constrained problems:

   $$\widehat{\beta}_\lambda \in \arg\min_\beta \{L(\beta) + \lambda P(\beta)\}, \qquad \lambda > 0,$$

   $$\min_\beta L(\beta) \quad \text{subject to} \quad P(\beta) \le t.$$

   Assume minimizers exist.

   (a) **Connect the two formulations.** Set $t = P(\widehat{\beta}_\lambda)$. Prove that $\widehat{\beta}_\lambda$ also solves the constrained problem. *Hint:* suppose a feasible $\widetilde{\beta}$ has strictly smaller loss. Compare the two penalized objectives. This question asks only for this direction, not for a one-to-one correspondence between λ and t.

   (b) **What does a larger penalty do?** For $0 < \lambda_1 < \lambda_2$, choose any corresponding minimizers $\widehat{\beta}_1, \widehat{\beta}_2$. Prove that $P(\widehat{\beta}_2) \le P(\widehat{\beta}_1)$. *Hint:* since $\widehat{\beta}_1$ minimizes the objective at λ_1, its objective is no larger than the value at $\widehat{\beta}_2$. Write this inequality, then the corresponding inequality at λ_2, and add them. Does this imply that every individual $|\widehat{\beta}_j|$ must decrease? Explain what the inequality does and does not establish.

   (c) **Interpret the geometry.** For $p = 2$ and $t > 0$, sketch the ridge and lasso constraint regions, labeling their axis intercepts. Add schematic elliptical squared-loss contours and explain why lasso can produce exactly zero coefficients more readily than ridge. Must every lasso fit contain a zero coefficient? Interpret λ as the penalty per unit of $P(\beta)$ and t as its upper limit. Here ridge uses $\|\beta\|^2 \le t$, so the radius is $\sqrt{t}$; the lecture's $\|\beta\| \le t$ instead has radius t.

2. **Distance from a point to a linear decision boundary.**

   To compare candidate boundaries in a support vector machine, we need the shortest distance from an observation to a boundary. Let

$$H = \{z \in \mathbb{R}^p : w^\top z + b = 0\}, \qquad w \in \mathbb{R}^p \setminus \{0\}, \quad b \in \mathbb{R},$$

   and let $x_i \in \mathbb{R}^p$ be a fixed point. Write $\|v\|_2 = \sqrt{v^\top v}$ for the Euclidean norm. Prove that

$$\operatorname{dist}(x_i, H) := \min_{z \in H} \|x_i - z\|_2 = \frac{|w^\top x_i + b|}{\|w\|_2}$$

   by completing the following steps.

   (a) For any two points $z_1, z_2 \in H$, show that $w^\top (z_2 - z_1) = 0$. Explain why this means that w is perpendicular to the boundary.

   (b) Find a scalar a such that $z_* = x_i - aw$ belongs to H. Show that the segment from z_* to x_i is perpendicular to H, and calculate $\|x_i - z_*\|_2$.

   (c) Prove that z_* is the closest point in H to x_i, rather than merely a point on the boundary. Conclude the stated distance formula. *Hint:* for an arbitrary $z \in H$, write $x_i - z = (x_i - z_*) + (z_* - z)$ and use part (a).

   (d) Replace (w, b) by (cw, cb), where $c > 0$. Show that the boundary and the distance are unchanged, even though the score $w^\top x_i + b$ is multiplied by c. Why should we divide the absolute score by $\|w\|_2$ when comparing distances to candidate boundaries?

3. **From maximum margin to convex optimization.**

Let (x_i, y_i), $i = 1, \ldots, n$, be training observations with $x_i \in \mathbb{R}^p$ and $y_i \in \{-1, +1\}$. Assume both labels occur and the data are strictly linearly separable: some (w, b) satisfies $y_i(w^\top x_i + b) > 0$ for every i. For $w \neq 0$, define the geometric margin

$$\gamma(w, b) = \min_i \frac{y_i(w^\top x_i + b)}{\|w\|_2}.$$

The geometric goal is to maximize $\gamma(w, b)$. We will show why the following convex quadratic program solves the same boundary-selection problem:

$$\min_{w,b} \frac{1}{2}\|w\|_2^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 \quad \text{for every } i. \tag{QP}$$

(a) **Why is the original problem not a convex minimization?** Consider $x_1 = (1, 0)^\top$, $y_1 = +1$, $x_2 = (-1, 0)^\top$, $y_2 = -1$, and restrict $b = 0$. Show that $\gamma(w, 0) = w_1/\sqrt{w_1^2 + w_2^2}$. With $u = (1, 0)^\top$ and $v = (1, 2)^\top$, calculate $\gamma(u, 0)$, $\gamma(v, 0)$, and $\gamma((u+v)/2, 0)$. Maximizing γ is the same as minimizing $h = -\gamma$. A convex function must satisfy $h((u + v)/2) \leq [h(u) + h(v)]/2$. Check that h violates this inequality, even though the whole segment between u, v has $w_1 > 0$.

(b) **Remove the arbitrary scale.** For any strictly separating (w, b), let $m = \min_i y_i(w^\top x_i + b) > 0$ and set $(\widetilde{w}, \widetilde{b}) = (w/m, b/m)$. Using Question 2(d), show that this changes neither the classification rule nor the geometric margin, while

$$\min_i y_i(\widetilde{w}^\top x_i + \widetilde{b}) = 1, \qquad \gamma(w, b) = \frac{1}{\|\widetilde{w}\|_2}.$$

(c) **Prove equivalence of the optimization problems.** Show that every feasible point of (QP) has $w \neq 0$, and that at any optimum of (QP), the smallest signed score equals 1. Then prove that a (QP) optimum maximizes geometric margin, and that normalizing any maximum-margin separator as in part (b) gives a (QP) optimum. If the optimal objective value of (QP) is J_*, express the maximum geometric margin γ_* in terms of J_*. *Hint:* if all signed scores exceed 1, shrink both w and b.

(d) **Explain the computational benefit.** Show that the allowed parameter region of (QP) is convex and that its objective is convex in (w, b). You may use that a region given by linear inequalities is convex and a sum of squares is convex. Does leaving the intercept b unpenalized destroy convexity? Explain why this reformulation changes the computational form but not the maximum-margin criterion.

4. **From slack variables to hinge loss.**

   For labels $y_i \in \{-1, +1\}$ and a fixed $C > 0$, consider

$$\begin{aligned} \min_{w,b,\xi_1,\ldots,\xi_n} \quad & \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n}\xi_i \\ \text{subject to} \quad & y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad i = 1, \ldots, n. \end{aligned}$$

   Show that it is equivalent to

$$\min_{w,b} \quad \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n}\max\{0, 1 - y_i(w^\top x_i + b)\}.$$

   Use only the two inequalities on each slack variable and $C > 0$.

   (a) Fix w, b and write $m_i = y_i(w^\top x_i + b)$. Express the two constraints on ξ_i as a single lower bound.

   (b) Find the minimizing ξ_i for these fixed w, b. Explain why the assumption $C > 0$ matters.

   (c) Substitute these optimal slacks into the objective. Explain why both problems have the same optimal value and the same optimal pairs (w, b), and how to recover the slacks from a solution of the hinge-loss problem.

5. **Why the SVM score is a weighted sum of inner products.**

Use the soft-margin problem in Question 4, with $C > 0$. The following function combines the objective and constraints using nonnegative numbers a_i, μ_i:

$$L = \frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i + \sum_i a_i[1 - \xi_i - y_i(w^\top x_i + b)] - \sum_i \mu_i \xi_i.$$

Take the following equations as given, without proving where they come from: at a solution, the derivatives with respect to w, b, ξ_i vanish, $a_i, \mu_i \geq 0$, and

$$a_i[1 - \xi_i - y_i(w^\top x_i + b)] = 0, \qquad \mu_i \xi_i = 0.$$

In each of the last two products, if one factor is positive, the other must be zero. That elementary observation and differentiation are all that you need for this exercise. You are not asked to derive these equations.

(a) Differentiate L with respect to w and show that $w = \sum_i a_i y_i x_i$. *Hint:* $\nabla_w(\|w\|_2^2/2) = w$ and $\nabla_w(w^\top x_i) = x_i$.

(b) Substitute this result into $f(x) = w^\top x + b$ to obtain $f(x) = b + \sum_i a_i y_i x_i^\top x$. Explain why a point with $a_i = 0$ contributes no term to this score.

(c) Differentiate with respect to ξ_i and use nonnegativity to prove $0 \leq a_i \leq C$. Must a_i be either 0 or 1? As a check, differentiating with respect to b should give $\sum_i a_i y_i = 0$.

(d) Write $m_i = y_i f(x_i)$. Use the given conditions and Question 4 to show: $m_i > 1$ implies $a_i = 0$; $0 < a_i < C$ implies $m_i = 1$; and $m_i < 1$ implies $a_i = C$. Points with $a_i > 0$ are called support vectors. Explain why soft-margin support vectors need not all lie on $m_i = 1$. *Hint:* first use $\xi_i = \max\{0, 1 - m_i\}$.

Optional extension: From linear SVM to kernels (ungraded)

This extension is not required and carries no marks or bonus points. Assume the data are strictly linearly separable and both labels occur. Start from the hard-margin problem in Question 3 and the following function:

$$L(w, b, a) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^{n} a_i[1 - y_i(w^\top x_i + b)], \qquad a_i \geq 0.$$

For fixed a, let $g(a)$ be the lowest value of L as w, b vary; write $g(a) = -\infty$ if L can decrease without bound. You may use this fact without proof: maximizing $g(a)$ over $a_i \geq 0$ gives an equivalent way to solve the hard-margin problem, with the same optimal objective value. Your task is to simplify this expression, not to prove the fact. At a finite minimum in w, b, their derivatives are zero.

(a) Differentiate L with respect to w and b. Show that a finite lowest value requires $\sum_i a_i y_i = 0$, and that its minimizing w is $w = \sum_i a_i y_i x_i$. *Hint:* if the coefficient of the unrestricted scalar b is nonzero, what happens when b tends to one of $+\infty$ or $-\infty$?

(b) Substitute this w into L and simplify $g(a)$. Write the resulting maximization problem, then negate its objective to obtain a minimization. *Hint:* put $s = \sum_i a_i y_i x_i$ and use $\|s\|^2 = \sum_{i,j} a_i a_j y_i y_j x_i^\top x_j$.

(c) Identify exactly where the training features appear in this problem. Show that its minimization objective is convex by writing the double sum as a squared norm. You may use that a squared norm of a linear function is convex and adding a linear term preserves convexity. Substitute the representation of w into $f(x) = w^\top x + b$ to obtain a prediction formula using only inner products.

(d) Replace $x_i^\top x_j$ by a kernel $K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$. Write the modified optimization problem and prediction formula. Contrast linear and RBF kernels, and explain why changing the kernel requires refitting the weights. You may assume the transformed data remain strictly separable for this hard-margin exercise. In the soft-margin version taught in class, the additional bounds are $a_i \leq C$; their derivation is not required.